Figure 8.12 The Black team implements its data product on top of Kafka.

The White team's data product sources data from an S3 bucket; the sourcing is done via a custom number-crunching application. The app uses the Kafka Producer API to write the datasets into a Kafka stream. A Kafka Connect sink is configured to dump this data into a Snowflake table as well. These two components, the Snowflake table and the Kafka streams port, become the data output ports.

The second data product from the White team is really simple in its architecture. It uses ksqlDB to connect to the existing event stream from the team's first data product, transforms the data using SQL, and then writes it again to an event stream that functions as the data output port. The data product is depicted in figure 8.13.

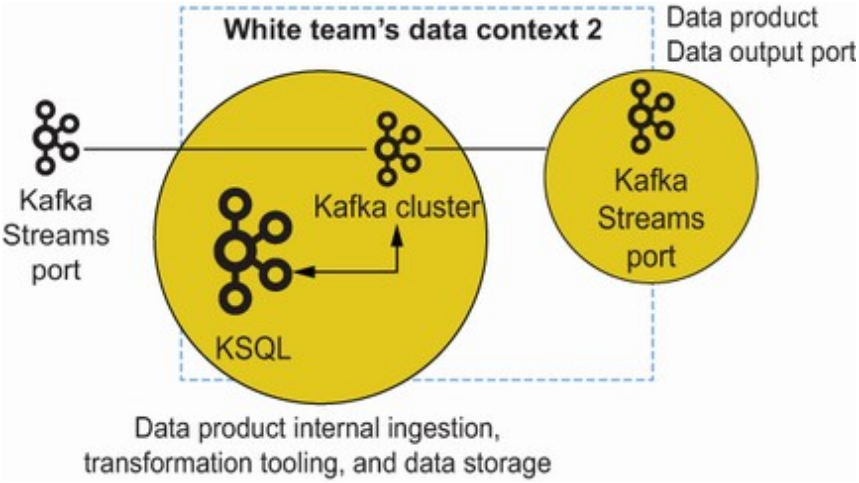


Figure 8.13 The White team uses ksqlDB to source another team's data to produce a second data product.

The recommendation engine in this case could just source the Kafka streams, but in our case, it is easier for it to take batch data that is put into the S3 bucket via a standard Kafka sink.

The business analysts are a bit special in this case. They have no direct way of consuming data products, not inside the Kafka ecosystem. For that, we utilize Trino as a query engine to query across all data products and allow